

Exercice 1

$$a) N = \left\lceil \frac{\log_{10}((10^{AP/10} - 1) / (10^{AS/10} - 1))}{2 \log \frac{\omega_p}{\omega_s}} \right\rceil$$

$$= \left\lceil \frac{\log_{10}((10^{\frac{7}{10}} - 1) / (10^{\frac{16}{10}} - 1))}{2 \log(\frac{0,2\pi}{0,3\pi})} \right\rceil$$

$$= \lceil 2,78 \rceil = \boxed{3}$$

$$\omega_p = \frac{\omega_p}{(10^{AP/10} - 1)^{\frac{1}{2N}}} = \frac{0,2\pi}{(10^{\frac{7}{10}} - 1)^{\frac{1}{6}}} \approx 0,5$$

$$\omega_s = \frac{\omega_s}{(10^{AS/10} - 1)^{\frac{1}{2N}}} = \frac{0,3\pi}{(10^{\frac{16}{10}} - 1)^{\frac{1}{6}}} \approx 0,5$$

$$\omega_c = \frac{\omega_p + \omega_s}{2} = \frac{0,5 + 0,5}{2} = \frac{1}{2} = 0,5$$

b) N est impair donc on a $k = 0, 1, \dots, 5$ et $\theta = \frac{k\pi}{N}$

Il n'y a pas de zéros pour ce filtre

$$P_0 \quad \theta = 0^0$$

$$P_1 \quad \theta = \frac{\pi}{3}$$

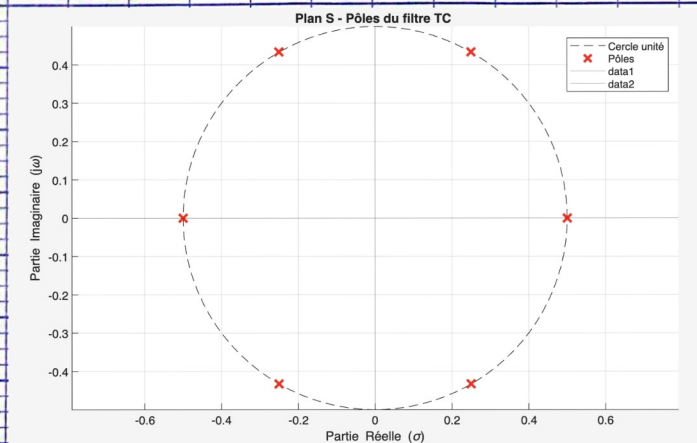
$$P_2 \quad \theta = \frac{2\pi}{3}$$

$$P_3 \quad \theta = \frac{3\pi}{3} = \pi$$

$$P_4 \quad \theta = \frac{4\pi}{3}$$

$$P_5 \quad \theta = \frac{5\pi}{3}$$

†



c) $H(S') = \frac{\omega_c^3}{(S' - P_2)(S' - P_3)(S' - P_4)}$, P_2, P_3, P_4 sont les pôles stables.
Calculons la valeur de ces pôles:

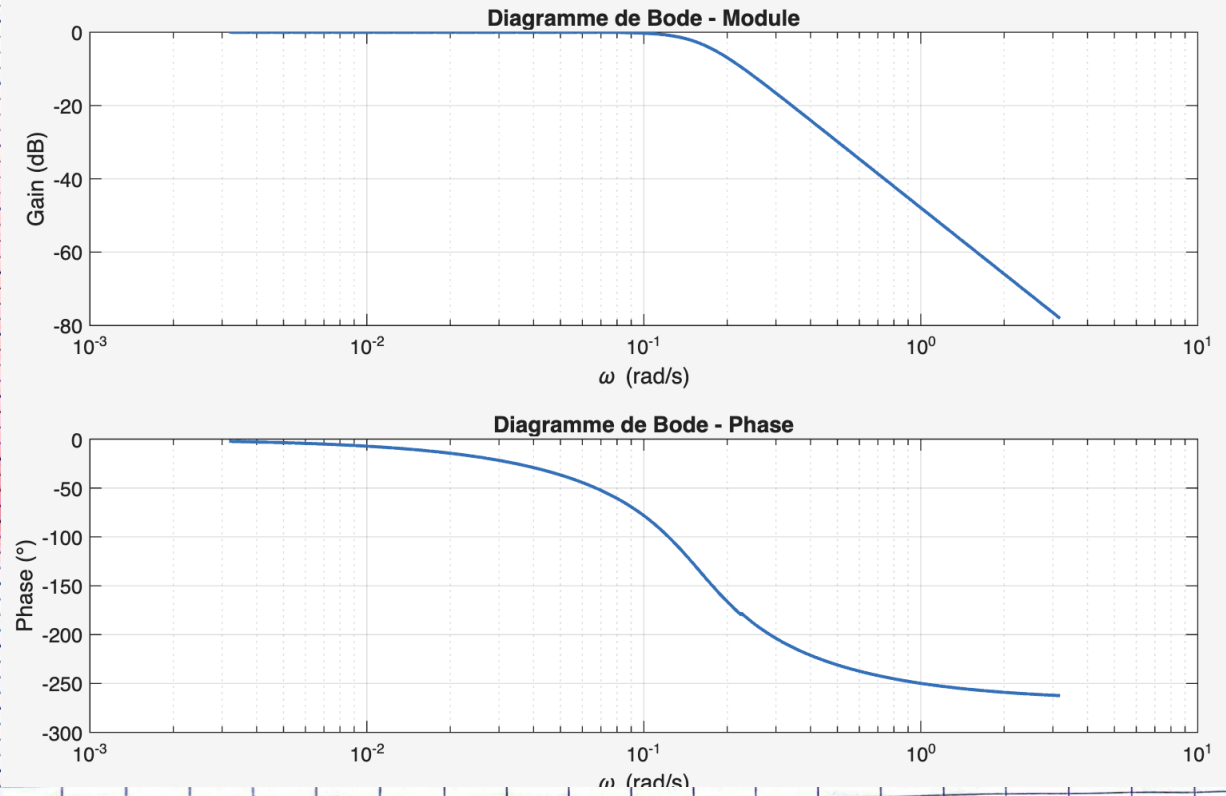
$$-P_2 = \omega_c e^{j2\pi/3} = \omega_c \left(\cos \frac{2\pi}{3} + j \sin \frac{2\pi}{3} \right) = -0,25 + 0,433j$$

$$-P_3 = \omega_c e^{j\pi} = \omega_c (\cos \pi + j \sin \pi) = -0,5$$

$$-P_4 = \omega_c e^{j4\pi/3} = \omega_c \left(\cos \frac{4\pi}{3} + j \sin \frac{4\pi}{3} \right) = -0,25 - 0,433j$$

$$\text{Ainsi } H(S') = \frac{0,5^3}{(S' + 0,25 - 0,433j)(S' + 0,5)(S' + 0,25 + 0,433j)}$$

d)



$$\begin{aligned}
 e) H(s) &= \frac{(0,5)^3}{(s+0,25-j0,433)(s+0,5)(s+0,25+j0,433)} \\
 &= \frac{(b_0)/s^3}{(s^3 + a_2s^2 + a_1s + a_0)} \\
 &= \frac{b_0/s^3}{1 + \frac{a_2}{s} + \frac{a_1}{s^2} + \frac{a_0}{s^3}}
 \end{aligned}$$

